**King Fahd University of Petroleum and Minerals**
**Department of Mathematics**
*Student effort not Official. Generated by Ice ❄. دعواتكم له*

# Math 208 Major Exam II Term 231

Original exam date: 7 November 2023. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

---

**1.** A linear homogeneous constant-coefficient differential equation which has the general solution

$$y(x) = e^{-4x}\left(c_1\cos(3x) + c_2\sin(3x)\right)$$

is

(a) $y'' - 6y' + 25y = 0$

(b) $y'' - 8y' - 25y = 0$

(c) $y'' + 6y' + 25y = 0$

(d) $y'' + 8y' + 25y = 0$

(e) $y'' - 8y' + 25y = 0$

**2.** If $y_p = A + Bx$ is a particular solution of the differential equation

$$y'' - y' - 2y = 3x + 4,$$

then $4A + 4B =$

(a) $-11$

(b) $12$

(c) $-10$

(d) $9$

(e) $0$

**3.** The general solution of the differential equation

$$4y'' - 12y' + 9y = 0$$

is

(a) $y = c_1e^{x} + c_2xe^{x}$

(b) $y = c_1e^{3x} + c_2e^{-3x}$

(c) $y = c_1e^{\frac{3}{2}x} + c_2xe^{\frac{3}{2}x}$

(d) $y = c_1e^{3x} + c_2xe^{3x}$

(e) $y = c_1e^{\frac{3}{2}x} + c_2xe^{-\frac{3}{2}x}$

**4.** The characteristic polynomial of the matrix $\begin{bmatrix} 3 & 5 & -2 \\ 0 & 2 & 0 \\ 0 & 2 & 1 \end{bmatrix}$ is $p(\lambda) =$

(a) $-\lambda^3 + 6\lambda^2 - 11\lambda + 6$

(b) $\lambda^3 - 6\lambda^2 + 9\lambda + 6$

(c) $\lambda^3 - 5\lambda^2 + 11\lambda + 6$

(d) $\lambda^3 + 4\lambda^2 - 2\lambda + 6$

(e) $\lambda^3 + 6\lambda^2 - 11\lambda + 4$

**5.** Which one of the following set of functions are linearly dependent

(a) $y_1(x) = 1,\ y_2(x) = x,\ y_3(x) = 2x + 3$

(b) $y_1(x) = 2,\ y_2(x) = 3x,\ y_3(x) = x^3$

(c) $y_1(x) = 1,\ y_2(x) = x,\ y_3(x) = x^2$

(d) $y_1(x) = x,\ y_2(x) = x^2 + x,\ y_3(x) = x^3 - 1$

(e) $y_1(x) = x,\ y_2(x) = x^2,\ y_3(x) = x^3$

**6.** The Wronskian of the functions

$$f(x) = 1, \quad g(x) = x, \quad h(x) = x^2$$

is

(a) 4

(b) 2

(c) 0

(d) 3

(e) $-4$

**7.** An appropriate form of a particular solution $y_p$ for the non-homogeneous differential equation $y'' + 4y = 3x\cos(2x)$ is given by $y_p(x) =$

(a) $(Ax^2 + Bx^3)\cos(2x) + (C + Dx)\sin(2x)$

(b) $(Ax + Bx^2)\cos(2x) + (Cx + Dx^2)\sin(2x)$

(c) $(Ax + Bx)\cos(2x) + (C + Dx)\sin(2x)$

(d) $(A + Bx)\cos(2x) + (Cx + Dx^2)\sin(2x)$

(e) $(Ax + Bx^2)\cos(2x) + (C + Dx)\sin(2x)$

**8.** Let the solution space of the system

$$\begin{aligned} x_1 - 2x_2 - 9x_3 + 7x_4 &= 0 \\ x_1 + x_2 + 3x_3 + 4x_4 &= 0 \\ x_1 + 4x_2 + 15x_3 + x_4 &= 0 \end{aligned}$$

has all linear combinations of the two vectors $u = (1, -4, 1, 0)$ and $v = (\alpha, 1, \beta, 1)$. Then $\alpha + \beta =$

(a) 3

(b) $-3$

(c) $-5$

(d) 5

(e) 0

**9.** Consider the subspace $S$ of $\mathbb{R}^4$ defined by $S = \{(a, b, c, d) \mid a = b + c + d\}$. A basis of $S$ is consisting of the vectors

(a) $v_1 = (1, 1, 1, 0),\ v_2 = (0, 1, 1, 1),\ v_3 = (0, 1, 1, 0)$

(b) $v_1 = (1, 1, 0, 0),\ v_2 = (1, 0, 1, 0),\ v_3 = (1, 0, 0, 1)$

(c) $v_1 = (1, 1, 1, 1),\ v_2 = (0, 0, 1, 0),\ v_3 = (1, 0, 0, 1)$

(d) $v_1 = (0, 1, 0, 0),\ v_2 = (0, 0, 1, 0),\ v_3 = (0, 0, 0, 1)$

(e) $v_1 = (1, 1, 0, 0),\ v_2 = (0, 0, 1, 1),\ v_3 = (0, 1, 0, 1)$

**10.** If the matrix $A = \begin{bmatrix} 5 & -3 \\ 2 & 0 \end{bmatrix}$ is diagonalizable with a diagonalizing matrix $P$ and a diagonal matrix $D$ such that $P^{-1}AP = D$, then

(a) $P = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix},\ D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

(b) $P = \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix},\ D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

(c) $P = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix},\ D = \begin{bmatrix} 0 & 2 \\ 3 & 0 \end{bmatrix}$

(d) $P = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix},\ D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

(e) $P = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix},\ D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

**11.** If the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 5 & -9 \\ 2 & 5 & m \end{bmatrix}$ is equal to 2, then $m =$

(a) 4

(b) $-5$

(c) 5

(d) 2

(e) 3

**12.** The general solution of the differential equation

$$y^{(3)} + 3y'' - 4y = 0$$

is

(a) $y = c_1 e^{x} + c_2 e^{3x} + c_3 x e^{3x}$

(b) $y = c_1 e^{x} + c_2 e^{2x} + c_3 x e^{2x}$

(c) $y = c_1 e^{x} + c_2 e^{-2x} + c_3 x e^{-2x}$

(d) $y = c_1 e^{-x} + c_2 e^{-2x} + c_3 x e^{-2x}$

(e) $y = c_1 e^{-x} + c_2 e^{2x} + c_3 x e^{2x}$

**13.** Given that $y_p = u_1(x)\cos(3x) + u_2(x)\sin(3x)$ is a particular solution of the differential equation

$$y'' + 9y = 2\sec(3x),$$

then $u_1(x) =$

(a) $\frac{1}{3}\ln|\cos(3x)|$

(b) $\frac{2}{9}\ln|\cos(3x)|$

(c) $\frac{1}{9}\ln|\cos(3x)|$

(d) $\frac{1}{7}\ln|\cos(3x)|$

(e) $\frac{2}{7}\ln|\cos(3x)|$

**14.** The rank of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & -9 & 9 \\ 3 & 6 & 2 & 1 & 6 \\ 2 & 4 & 1 & 2 & 3 \\ 4 & 8 & 3 & 0 & 9 \end{bmatrix}$$

is

(a) 1

(b) 5

(c) 3

(d) 4

(e) 2

**15.** The eigenvector associated with the eigenvalue $\lambda = -1$ of the matrix $A = \begin{bmatrix} 7 & -8 \\ 6 & -7 \end{bmatrix}$ is $\begin{bmatrix} a \\ 1 \end{bmatrix}$, where $a =$

(a) $-2$

(b) 0

(c) $-1$

(d) 1

(e) 2

**16.** The solution of the initial-value problem

$$y'' - y = 0, \quad y(0) = 0 \text{ and } y'(0) = 5$$

is

(a) $y = e^x - e^{-x}$

(b) $y = e^{-x} - e^x$

(c) $y = \frac{3}{2}e^x - \frac{3}{2}e^{-x}$

(d) $y = -\frac{5}{2}e^x + \frac{5}{2}e^{-x}$

(e) $y = \frac{5}{2}e^x - \frac{5}{2}e^{-x}$

**17.** If the characteristic polynomial of the matrix $A = \begin{bmatrix} 5 & -6 & 3 \\ 6 & -7 & 3 \\ 6 & -6 & 2 \end{bmatrix}$ is $p(\lambda) = -(\lambda+1)^2(\lambda-2)$, then a basis for the eigenspace of $\lambda = -1$ is $v_1 = \begin{bmatrix} m \\ 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} -1 \\ 0 \\ n \end{bmatrix}$, where $m+n =$

(a) 4

(b) 5

(c) 0

(d) 1

(e) 3

**18.** The general solution of the differential equation

$$y^{(4)} + 18y'' + 81y = 0$$

is

(a) $y = (c_1 + c_2x)e^x\cos(2x) + (c_3 + c_4x)e^x\sin(2x)$

(b) $y = (c_1 + c_2x)e^x\cos(3x) + (c_3 + c_4x)e^x\sin(3x)$

(c) $y = c_1\cos(3x) + c_2\sin(3x) + c_3\cos(2x) + c_4\sin(2x)$

(d) $y = (c_1 + c_2x)\cos(3x) + (c_3 + c_4x)\sin(3x)$

(e) $y = (c_1 + c_2x)\cos(2x) + (c_3 + c_4x)\sin(2x)$

**19.** Consider the vectors

$$v_1 = (-1, 2, 3),\ v_2 = (3, 1, -2),\ v_3 = (2, 3, 0) \text{ and } w = (1, 5, 8).$$

If $w = av_1 + bv_2 + cv_3$, then $a + b - c =$

(a) 17

(b) 15

(c) 19

(d) 13

(e) 21

**20.** A value of $k$ for which the following vectors

$$v_1 = (k, 0, k), \quad v_2 = (2k, -3, 4) \quad \text{and} \quad v_3 = (3, 5, 2)$$

are linearly dependent is

(a) $\dfrac{-16}{5}$

(b) $\dfrac{7}{5}$

(c) $\dfrac{10}{5}$

(d) $\dfrac{17}{10}$

(e) $\dfrac{-8}{5}$

# Answer Key

*Student effort not Official. Generated by Ice ❄. دعواتكم له*

| Q | Ans | Brief explanation |
|---|---|---|
| 1 | **(d)** | Roots $-4 \pm 3i$: $(r+4)^2+9 = r^2+8r+25$, so $y''+8y'+25y=0$. |
| 2 | **(a)** | $-2B=3$, $-B-2A=4$ give $B=-\frac{3}{2}$, $A=-\frac{5}{4}$, so $4A+4B=-11$. |
| 3 | **(c)** | $(2r-3)^2=0$, double root $r=\frac{3}{2}$: $y=c_1e^{3x/2}+c_2xe^{3x/2}$. |
| 4 | **(a)** | $\det(A-\lambda I)=(3-\lambda)(2-\lambda)(1-\lambda)=-\lambda^3+6\lambda^2-11\lambda+6$. |
| 5 | **(a)** | $2x+3=2\cdot x+3\cdot 1$, so $\{1,x,2x+3\}$ is dependent; the others are independent. |
| 6 | **(b)** | $W=\begin{vmatrix}1 & x & x^2\\ 0 & 1 & 2x\\ 0 & 0 & 2\end{vmatrix}=2$. |
| 7 | **(b)** | $y_c$ has $\cos 2x, \sin 2x$, so multiply the trial form by $x$: $(Ax+Bx^2)\cos 2x+(Cx+Dx^2)\sin 2x$. |
| 8 | **(c)** | Reduces to $x_2+4x_3-x_4=0$; basis $(1,-4,1,0)$, $(-5,1,0,1)$, so $\alpha+\beta=-5$. |
| 9 | **(b)** | Free $b,c,d$: $(b+c+d,b,c,d)=b(1,1,0,0)+c(1,0,1,0)+d(1,0,0,1)$. |
| 10 | **(a)** | $\lambda=2,3$ with eigenvectors $(1,1),(3,2)$: $P=\begin{bmatrix}1 & 3\\ 1 & 2\end{bmatrix}$, $D=\operatorname{diag}(2,3)$. |
| 11 | **(d)** | $\det=3m-6=0$, so $m=2$ (rows 1 and 2 are independent, rank 2). |
| 12 | **(c)** | $r^3+3r^2-4=(r-1)(r+2)^2$: $y=c_1e^x+c_2e^{-2x}+c_3xe^{-2x}$. |
| 13 | **(b)** | $W=3$, $u_1'=-\frac{2}{3}\tan 3x$, so $u_1=\frac{2}{9}\ln\lvert\cos 3x\rvert$. |
| 14 | **(e)** | After clearing column 1 the other rows are multiples of $(0,0,1,-4,3)$: rank 2. |
| 15 | **(d)** | $A+I$ gives $x=y$, so $\mathbf{v}=(1,1)$ and $a=1$. |
| 16 | **(e)** | $r=\pm 1$; $c_1+c_2=0$, $c_1-c_2=5$ give $y=\frac{5}{2}e^x-\frac{5}{2}e^{-x}$. |
| 17 | **(e)** | $A+I$ gives $2x-2y+z=0$; $m=1$, $n=2$, so $m+n=3$. |
| 18 | **(d)** | $(r^2+9)^2=0$: $r=\pm 3i$ double, so $y=(c_1+c_2x)\cos 3x+(c_3+c_4x)\sin 3x$. |
| 19 | **(b)** | Solve the system: $a=6,\ b=5,\ c=-4$, so $a+b-c=15$. |
| 20 | **(d)** | $\det=10k^2-17k=k(10k-17)=0$, so $k=\frac{17}{10}$ (the only listed value). |